Test 2

Up to the reading for chapter 9

Reading 6: Next Wednesday

(Mostly Introspective)

(Will include Chapter 10 reading)

Assignment 5:

Due next Sunday.

Chapter 8 Recurrence Equations

Warmup by example

Suppose

$a_n = a_{n-1} + 6a_{n-2}$ when $n \geq 2$ (This is the recurrence equation)

$a_0 = 1$ and $a_1 = 5$ (These are initial values)

Goal:

To find a forumla for directly computing a_n

This sequence starts
$a_0 = 1$
$a_1 = 5 a_2 = a_1 + 6a_0 = 5 + 6 = 11$
$a_3 = a_2 + 6a_1 = 41$
$a_4 = a_3 + 6a_2 = 107$
etc

Strategy:

- Ignore the initial conditions (Until the end)
- Find all of the solutions to the recurrence (With parameters)
- Solve for the paramters to match the initial conditions
- To find all solutions:
  - Look for really nice solutions
  - Include nice solutions together

Guess

$b_n = 3^n$ might be nice

$b_{n-1} + 6b_{n-2}$ is then
$3^{n-1} + 6 * 3^{n-2}$
$= 3 * 3^{n-2} + 6 * 3^{n-2}$
$= 9 * 3^{n-2} = 3^n = b_n$

so for all $n \geq 2$

$b_n = b_{n-1} + 6b_{n-2}$

Rewrite reecurrence as

$a_n - a_{n-1} - 6a_{n-2} = 0$

Could there be another nice solution?

so:

$c_n = y^n$ we just saw directly, that $y = 3$ worked.

If there is, then it works when $n = 2$, so $y^2 - y - 6 = 0$

We can solve quadratic equations:

1. By inspection
2. By quadratic forumla

$y^2 - y - 6 = (y-3)(y+2)$ so

y = 3 y = −2

At this point

$b_n = 3^n$ and $c_n = (-2)^n$

are both “special” solutions to the recurrence.

But so are their linear combinations,

If $d_n = \alpha b_n + \beta c_n$

then

$= \alpha(b_{n-1} + 6b_{n-2}) + \beta(c_{n-1} + 6c_{n-2})$

$= \alpha b_{n-1} + \beta c_{n-2} + 6(\alpha b_{n-2} + \beta c_{n-2})$

$d_{n-1} + 6d_{n-2}$

so d_n also satisfies the recurrence

$d_n = \alpha 3^n + \beta(-2)^n$ describes all possible solutions (Weca can solve for $\alpha\beta$ so $d_0 = a_0$ and $d_1 = a_1$

and then the structural induction says a and d continue to match)

In this case we want

$1 = d_0 = \alpha * 3^0 + b(-2)^0 = \alpha + \beta$

$5 = d_1 = 3\alpha - 2\beta$

$\alpha + \beta = 1$

$3\alpha - 2\beta = 5$

Two equations in two unknowns.

Subbing

$\alpha = 1 - b$ in equation two gives

$3(1-b) - 2b = 5$

$-5\beta = 2$

$\beta = -\frac{2}{5}$

thus

$\alpha = 1 - \left(-\frac{2}{5}\right) = \frac{7}{5}$

and our solution is:

$a_n = \frac{7}{5}3^n - \frac{2}{5}(-2)^n$

In general

To solve for

$a_n = \alpha_1 a_{n-1} + \alpha_2 a_{n-2} + \alpha_3 a_{n-3} + ... + \alpha_k a_{n-k}$

subject to initial conditions.

Try to find special solutions,

$b_n = y^n$

which must be solutions to the polynomial

$x^k - \alpha_1 x^{k-1} - \alpha_2 x^{k-2} - ... - \alpha_k = 0$ (This is a characteristic polynomial.)

If the Characteristic Polynomial has distinct roots and

Characteristic Polynomial $= (x - r_1)(x - r_2)...(x - r_k)$

then a generic soltuion ignoring initial conditions is

$a_n = c_1 r_1^n + c_2 r_2^n + ...c_k r_k^n$

We can solve for the k unknowns to match any arbitrary k initial conditions.

If there are repeated roots, we don't get enough parameters, and need different solutions.

Example

Solve $b_n = 5b_{n-1} - 6b_{n-2}$

subject to $b_0 = 7$ and $b_1 = 19$

$r^2 - 5r + 6 = 0$

$(r - 3)(r - 2)$

$r_1 = 2$
$r_2 = 3$

$7 = A + B$ = $A = 7 - B$
$19 = A2 + B3$

$19 = 2(7 - B) + 3B$

$19 = 14 - 2B + 3B$

$5 = B$

$A = 2$

$b_n = 2 * 2^n + 5 * 3^n$

What could go wrong

- Repeated Roots

Try solving

$c_n = 10c_{n-1} - 25c_{n-2}$

$c_0 = 7$ and $c_1 = 4$ $r^2 - 10r + 25$

$(r - 5)^2 = 0$

$r_0 = 5$

Fortunately $d_n = n5^n$

$10d_{n-1} - 25d_{n-2}$
$= 10(n-1)5^{n-1} - 25(n-2)5^{n-2}$
$= (2n-2)5^n - (n-2)5^n$
$= n5^n = d_n$

In this case, a generic soltuion (ignoring initial conditions) is

$c_n = \alpha 5^n + \beta n 5^n$

now solving

$c_0 = \alpha = 7$

$c_1 = 5\alpha + 5\beta = 4$

$35 + 5\beta = 4$ or $5\beta - 31$ or $\beta = -\frac{31}{5}$

The full solution in this case, is

$c_n = 7 * 5^n - \frac{31}{5} n 5^n$